

MODULE 4.1: CYBER FORENSICS

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Introduction to Cyber Forensics

- Cyber Forensics (also known as Computer Forensics or Digital Forensics) is the process of **identifying, collecting, preserving, analyzing, and presenting digital evidence** in a legally acceptable manner.
- Its goal is to investigate cybercrimes, recover data, and support legal proceedings.

Scope:

- Investigating cybercrimes such as hacking, financial fraud, identity theft, cyberbullying, and ransomware
- Recovering deleted, hidden, or encrypted files
- Analyzing devices like computers, mobiles, servers, IoT devices, and cloud systems
- Examining logs, emails, digital documents, and network traffic
- Presenting evidence in court with proper chain of custody

Cyber forensics plays a vital role in law enforcement, corporate investigations, and incident response.

Importance

- Helps identify offenders behind cybercrimes
- Provides **legally valid evidence** for courts
- Supports organizations in detecting security breaches
- Recovers lost or damaged data after attacks
- Ensures the integrity and authenticity of digital evidence
- Helps prevent future incidents through root-cause analysis

Applications

- **Criminal investigations:** Cyber fraud, child exploitation, cyberstalking, theft
- **Corporate investigations:** Data breaches, employee misuse, intellectual property theft
- **Incident response:** Analyzing malware, identifying attack vectors
- **Legal and civil cases:** Evidence of digital contracts, emails, or communication
- **Network forensics:** Monitoring network traffic to detect intrusions
- **Mobile forensics:** Analyzing smartphones, apps, GPS, and call logs

Digital Evidence and Its Types

Digital evidence refers to **any information stored or transmitted electronically** that can be used in an investigation or legal case. It must be authentic, reliable, complete, and preserved without tampering. **Types of Digital Evidence**

1. **Real-time/Volatile Evidence:** Data that disappears when power is lost. Examples: RAM contents, active network connections, running processes, temporary files
2. **Non-volatile Evidence:** Data stored permanently on devices. Examples: files on hard disks, SSDs, USB drives, logs, emails
3. **Metadata Evidence:** Information about files or activities. Examples: timestamps, IP addresses, geolocation, sender information
4. **Network Evidence:** Logs, packets, server records, and communication traces
5. **Multimedia Evidence:** Images, videos, audio recordings used in cyber or criminal cases
6. **Cloud Evidence:** Data stored on remote servers, SaaS platforms, and cloud logs

Digital evidence must be handled with strict procedures to ensure it is accepted in court.

Digital Forensic Lifecycle

The Digital Forensic Lifecycle describes the **systematic process** followed during a forensic investigation to ensure accuracy and legal validity.

1. **Identification:** Recognizing potential sources of evidence such as computers, phones, logs, emails, or cloud sources. Investigators decide what data is relevant to the case.
2. **Preservation:** Protecting the evidence to avoid tampering or loss. This includes isolating devices, taking forensic images, write-blocking storage, and securing hash values.
3. **Collection:** Acquiring data from identified sources using approved forensic tools and methods. This step must follow chain-of-custody rules to maintain legal integrity.
4. **Examination:** Extracting, recovering, and filtering relevant information. Deleted files, logs, browsing history, and hidden data are examined.
5. **Analysis:** Interpreting evidence to understand what happened, how it happened, and who was responsible. Investigators correlate timelines, user actions, system logs, and communication data.
6. **Documentation:** Recording every action taken during the investigation, including tools used, procedures followed, and findings gathered.
7. **Reporting & Presentation:** Preparing a comprehensive forensic report with clear, legally defensible findings. Investigators may present evidence in court and explain technical details in simple terms.

Point	Physical Evidence	Digital Evidence
Nature	Tangible, can be touched (weapons, fingerprints)	Intangible, stored electronically (files, logs)
Durability	Usually stable over time	Easily altered, deleted, or damaged
Collection	Collected from crime scenes using physical tools	Collected from devices, networks, and storage media
Visibility	Visible to the naked eye	Often hidden or requires tools to access
Preservation	Requires sealing & physical storage	Requires imaging, hashing, and digital preservation
Authentication	Proven using physical characteristics	Proven using hashes, timestamps, metadata
Volatility	Rarely volatile	Can be volatile (RAM, temp files)

Point	Computer Security	Digital Forensics
Primary Goal	Protect systems from attacks	Investigate incidents after they occur
Focus Area	Prevention and defense	Detection, analysis, and evidence collection
Timeframe	Works before an incident	Works after an incident
Activities	Firewalls, patching, monitoring, access control	Imaging drives, analyzing logs, recovering files
Outcome	Reduce vulnerabilities and threats	Provide evidence for legal or internal investigation
Approach	Proactive	Reactive
Tools Used	Antivirus, IDS, SIEM, firewalls	EnCase, FTK, Autopsy, disk imagers

Point	Forensic Policy	Security Policy
Purpose	Defines rules for investigation and evidence handling	Defines rules for protecting systems and data
Focus	Collection, preservation, analysis of digital evidence	Access control, network protection, user behavior
Used By	Forensic investigators, legal teams	IT security teams, employees, management
Scope	Covers incident response, chain of custody, reporting	Covers passwords, firewalls, acceptable use, encryption
Objective	Ensure evidence is valid and legally admissible	Prevent cyber attacks and minimize risk
Nature	Reactive — activated after incidents	Proactive — enforced continuously

Evidence Management

Types of Evidence

1. **Real (Physical) Evidence:** Tangible objects such as computers, hard drives, USB devices, CDs, phones, routers, etc. These items physically store or contain digital evidence.
2. **Digital Evidence:** Electronic information stored or transmitted using computers, networks, cloud, or mobile devices. Examples: emails, logs, documents, images, deleted files, browser history.
3. **Volatile Evidence:** Data stored in temporary memory that disappears when power is lost. Examples: RAM contents, active network connections, running processes. Must be collected **first** during investigations.
4. **Non-Volatile Evidence:** Data stored in permanent media like HDDs, SSDs, USBs, databases, and cloud storage. It remains even after shutdown.
5. **Documentary Evidence:** Digital records that can be used in court, such as emails, text messages, call logs, PDFs, screenshots, and system logs.
6. **Testimonial Evidence:** Statements by witnesses or digital forensics experts during investigation or trial.
7. **Demonstrative Evidence:** Visual presentations used to explain findings — charts, diagrams, timelines, recreated attack flows.

Rules of Evidence

Evidence must follow strict rules to be accepted in court. The main rules are:

1. **Admissibility:** Evidence must be obtained legally and must be relevant to the case.
2. **Authenticity:** Investigators must prove that the evidence is genuine and has not been altered. Hash values (MD5, SHA-256) are used to prove integrity.
3. **Reliability:** The method used to collect and analyze evidence must be scientifically valid and repeatable.
4. **Integrity / Chain of Custody:** A detailed record showing **who handled the evidence, when, and why**. This ensures evidence was not tampered with.
5. **Completeness:** Evidence must include both incriminating and exculpatory information.
6. **Preservation:** Evidence must be stored properly to prevent modification or loss.

Evidence Act (Legal Foundations)

The Evidence Act (such as **Indian Evidence Act Section 65B**, or similar laws in other countries) defines the legal rules for admitting electronic evidence in court. Key legal foundations include:

- **Electronic evidence is admissible** if proper procedures are followed.
- **Section 65B certificate** is needed to verify electronic records.
- Evidence must be obtained without violating privacy laws or cyber laws.
- Digital evidence must maintain **authenticity, integrity, and chain of custody**.
- Tools and methods used must be legally recognized.

These laws ensure that digital evidence is treated with the same seriousness as physical evidence.

Guidelines / Steps to Collect Digital Evidence

Collecting digital evidence must follow standardized forensic procedures to ensure accuracy and legal validity.

1. **Secure the Scene:** Ensure devices are not tampered with. Disconnect unauthorized persons from the scene.
2. **Identify Potential Evidence:** identify all devices: computers, mobiles, USB drives, routers, servers, cloud accounts, logs.
3. **Preserve Evidence:** Prevent modification by:
 - Using write blockers
 - Photographing the scene
 - Documenting device condition
 - Calculating hash values
4. **Collect Data Properly:** Take forensic images (bit-by-bit copies) of storage devices. Capture volatile data first (RAM, network connections). Use approved forensic tools (FTK Imager, EnCase, Autopsy).
5. **Maintain Chain of Custody:** Record who collected evidence, the time, location, and every person who handled it.
6. **Transport and Store Securely:** Store evidence in tamper-proof bags and secure lockers. Maintain logs for access.
7. **Document Everything:** Every step — identification, imaging, hash values, tools used, observations — must be documented for legal presentation.

Computer Equipments: Hardware, Storage Media & Data

Forensic investigators must understand key hardware components because evidence often resides inside or interacts with them. **Key Hardware Components**

- **Motherboard:** Connects all system components; investigators may check BIOS/UEFI settings for startup logs or boot order changes.
- **CPU:** Processes instructions; forensic tools often analyze CPU usage logs or system events.
- **RAM:** Stores volatile data like running processes, network connections, and encryption keys — must be captured quickly before shutdown.
- **Hard Drives / SSDs:** Primary storage for files, logs, deleted data, and system artifacts.
- **Network Interfaces:** Logs related to internet activity, MAC addresses, and communication trails.
- **Peripherals:** Keyboards, cameras, printers — may store useful logs or cached documents.
- **Mobile Devices:** Phones and tablets contain calls, messages, app data, GPS logs, and media files.
- **External Devices:** USB drives, external HDDs, and memory cards often store copied or stolen data.

Understanding hardware helps investigators decide what evidence to collect and how to handle each device without altering data.

Storage Devices

Digital evidence commonly resides on various storage media, each with different properties.

1. Hard Disk Drives (HDDs)

- Magnetic storage
- Data remnants often remain even after deletion (good for recoveries)
- Forensic tools can recover deleted partitions and hidden data

2. Solid State Drives (SSDs)

- Flash-based storage
- Faster but harder forensics due to TRIM command, which wipes deleted data
- Requires specialized tools and imaging techniques

3. USB Flash Drives

- Portable storage used for transferring data
- Often used for data exfiltration (stealing data)
- May contain hidden files, portable malware, or encrypted containers

4. Memory Cards (SD, microSD)

- Popular in cameras, phones, and IoT devices
- Store media, app data, configuration files
- Often used in mobile and multimedia forensic investigations

Each device must be handled with write blockers to prevent accidental modification.

Volatile vs Non-Volatile Data

Volatile Data: Volatile data is stored in memory components that lose data when power is removed. Volatile data must be captured **first** during a live response because once the system shuts down, it disappears permanently. Examples:

- RAM contents
- Running processes
- Network connections
- Encryption keys
- System caches.

Non-Volatile Data

Non-volatile data persists even after power loss. This data can be imaged and analyzed later using forensic tools Examples:

- Files on HDDs and SSDs
- USB drives and memory cards
- Logs, registry entries
- Emails, documents, images.

Why the distinction matters: Volatile data gives real-time insight into what happened during an incident, while non-volatile data provides long-term records and evidence for deeper analysis.

Role of the Forensics Investigator

A forensic investigator is responsible for **identifying, collecting, preserving, examining, and analyzing digital evidence** while maintaining strict legal and technical standards. Key duties include:

- Securing the crime scene and preventing tampering
- Collecting digital evidence using approved forensic methods
- Using write blockers and creating forensic images (bit-by-bit copies)
- Recovering deleted or hidden data
- Analyzing logs, network traffic, file systems, and suspicious activity
- Maintaining chain of custody documentation
- Preparing technical forensic reports
- Presenting findings in court in an accurate and unbiased manner

Investigators must be thorough, methodical, and legally compliant.

Legal Considerations

Forensic investigators must follow laws related to:

- Privacy and data protection
- Cybercrime laws
- Electronic Evidence Acts (e.g., Section 65B in India)
- Search warrants and seizure procedures
- Admissibility of digital evidence

Any violation can make the evidence invalid in court.

Chain of Custody

Chain of custody is the **chronological documentation** that shows who collected, handled, transferred, and stored the evidence. It ensures:

- Evidence is authentic
- Evidence was not altered
- Every handler is accountable

A proper chain of custody includes timestamps, signatures, device details, hash values, and storage conditions.

Ethical Requirements

Investigators must:

- Stay impartial and unbiased
- Avoid tampering or altering evidence
- Respect privacy and confidentiality
- Work only within legal permissions and scope
- Use approved forensic tools and methods
- Maintain transparency in all actions

Ethics protect the integrity of both the investigation and the investigator.

Documentation Practices

Everything must be documented, including:

- Tools used
- Commands executed
- Location and condition of devices
- Steps taken during acquisition and analysis
- Hash values before and after imaging
- Findings, timelines, and conclusions

Detailed documentation ensures the investigation is repeatable and defensible in court.

File Carving

- File carving is a technique used to **recover files from unallocated space, deleted partitions, or damaged storage** without relying on file system metadata.
- It identifies file headers and footers (signatures) to reconstruct documents, images, videos, or other data types.
- Example: Recovering a deleted JPG by finding its header (FFD8) and footer (FFD9).
- File carving is essential when file system structures are corrupted or intentionally destroyed.

Anti-Forensics Techniques

Anti-forensics refers to methods attackers use to **hide, alter, or destroy evidence** to avoid detection.

Common Anti-Forensics Methods

- **Data wiping / secure deletion:** Overwriting data to prevent recovery
- **Encryption:** Making data inaccessible without keys
- **Steganography:** Hiding data inside images/audio
- **Timestamp modification:** Changing file creation or access times
- **Log tampering:** Clearing logs or modifying entries
- **Rootkits:** Hiding processes, files, or system changes

Investigators must understand these methods to detect manipulation attempts.

Data Mining in Cyber Forensics

Data mining helps investigators analyze **large volumes of digital evidence** and extract meaningful patterns. **Uses in Forensics**

- Identifying suspicious user activity or anomalies
- Detecting fraud patterns or repeated behavior
- Correlating logs, emails, and network traffic
- Clustering related evidence across different devices
- Building timelines of events from large datasets

Techniques like classification, clustering, pattern recognition, and anomaly detection support forensic decision-making.

Seven Steps of Forensic Investigation

1. **Identification:** This is the initial phase where the investigator determines that a digital incident has occurred and identifies all potential sources of evidence.
 - **Goal:** To recognize the scope of the incident and locate the affected systems.
 - **Activity:** Defining the type of crime or incident (e.g., unauthorized access, data deletion) and listing all relevant electronic devices, cloud accounts, and data logs that need to be examined.
 - Examples include:
 - Computers, laptops, servers, mobile phones
 - USB drives, SD cards, hard disks
 - Emails, log files, browsing activity
 - Network traffic, router logs, firewall logs
 - Cloud storage accounts
 - Social media accounts or messaging apps
2. **Preservation:** Once identified, the evidence must be secured to prevent any changes or destruction, which could invalidate its legal standing.
 - **Goal:** To freeze the state of the evidence and maintain its integrity.
 - **Activity:** **Isolating the device** from the network, photographing the physical scene, and ensuring no one writes to the original media. This critical step precedes collection.
 - **Imaging:** Forensic imaging creates a **bit-by-bit copy** of the storage device. This ensures investigators analyze the copy while preserving the original evidence untouched. Tools used: FTK Imager, EnCase, dd, Guymager.
3. **Collection:** This step involves acquiring a forensic image of the digital media while strictly maintaining integrity and documenting the process.
 - **Goal:** To create a **forensically sound, bit-by-bit copy (image)** of the original storage media (hard drive, memory, etc.).
 - **Activity:** Using a **write-blocker** to ensure the original media is not altered. Calculating and recording the **cryptographic hash** (MD5 or SHA) of both the original media and the forensic image. The hash values must match to prove integrity. The **Chain of Custody** is formally established here.
 - **Common techniques include:**
 - **Live acquisition:** Collecting RAM, volatile data, network connections.
 - **Dead acquisition:** Imaging storage after the device is powered off.
 - **Remote acquisition:** Collecting evidence from remote or cloud systems.
 - **Logical acquisition:** Copying visible files and folders.
 - **Physical acquisition:** Copying entire storage media including deleted space.
4. **Examination:** The acquired evidence image is searched and filtered using specialized forensic tools to isolate relevant data.
 - **Goal:** To systematically search and filter the massive amount of recovered data for items relevant to the case.
 - **Activity:** Searching for keywords, recovering deleted files (file carving), and filtering out known operating system files to reduce the data set.
5. **Analysis:** The filtered data is interpreted to reconstruct the events that took place and draw logical conclusions.
 - **Goal:** To interpret the evidence, reconstruct the timeline, and link digital artifacts to specific actions or users.

- **Activity:** Correlating timestamps, analyzing metadata (e.g., when a document was last accessed), examining registry entries, and identifying the root cause and method of intrusion.
6. **Documentation:** This is an ongoing step performed throughout the entire investigation to record all processes and findings meticulously.
 - **Goal:** To create a transparent and verifiable record of the entire investigation.
 - **Activity:** Maintaining a detailed **log** of every tool used, every action taken, hash values, photographic evidence, and the location of physical and digital evidence. This documentation supports the final report.
 7. **Reporting:** The investigation culminates in a formal report that communicates the process and the findings to clients, management, or legal authorities.
 - **Goal:** To present the facts, methodologies, and conclusions clearly and accurately for legal or management review.
 - **Activity:** Writing a formal report that includes the scope of the investigation, the methods used, a list of critical evidence, and the final expert conclusion. The examiner must be prepared to testify to the findings as an **expert witness**.

Collecting Network-Based Evidence

Network forensics focuses on capturing, analyzing, and interpreting data that travels across networks. This evidence helps investigators trace attacks, identify intrusion methods, and reconstruct communication patterns.

Types of Network Evidence

Network-based evidence comes from various sources that record or transmit network activity.

1. **Packet Captures (PCAP Files):** Raw network packets collected from network interfaces. Contain source/destination IP, ports, protocols, payloads, timestamps, and more.
2. **Network Device Logs:** Routers, firewalls, switches, and IDS/IPS generate logs that record traffic, blocked attempts, configuration changes, and suspicious activities.
3. **Server and Application Logs:** Web server logs, DNS logs, email logs, database access logs, and authentication logs help track communication and user actions.
4. **Flow Records (NetFlow / IPFIX):** Summaries of network traffic (not full packets). Useful for analyzing large-scale attack patterns.
5. **Wireless Network Evidence:** Wi-Fi access logs, device MAC addresses, connection timestamps, and security settings.
6. **Cloud and VPN Logs:** Evidence from cloud platforms, VPN gateways, and remote access services. Each type helps reconstruct attacker behavior and timeline of events.

Capturing Network Traffic (PCAP, Logs)

Capturing network evidence requires tools and methods that don't alter the data.

Packet Capture (PCAP): Packet capture records complete network packets. Investigators use tools like **Wireshark**, **tcpdump**, or **tshark** to capture and analyze traffic. Packet captures can reveal:

- Suspicious connections
- Malware communication
- Credential theft
- Data exfiltration attempts

- MITM or spoofing attacks

Log Collection: Logs are collected from:

- Firewalls (blocked/allowed traffic)
- Routers/switches (routing entries, ARP tables)
- Web servers (requests, client IPs)
- Email servers (headers, delivery logs)
- DNS servers (queries, responses)

Logs help in identifying IP sources, attack times, unauthorized access attempts, and malicious requests.

Preservation: Captured packets and logs must be hashed and stored securely to maintain integrity.

Investigating Network Attacks

Network attacks often leave clear traces in network data. Investigators analyze this evidence to understand what happened. **Key Steps in Investigation:**

1. **Identify abnormal traffic:** Unusual spikes, repeated requests, non-standard ports, or unknown IPs.
2. **Trace attacker origin:** Examine source IPs, VPN connections, or compromised internal hosts.
3. **Reconstruct events:** Follow the sequence of packets to recreate the attack path (e.g., scanning → exploitation → data theft).
4. **Check for intrusion indicators:** Signs include port scans, failed login attempts, payload injections, DNS anomalies, or exfiltration patterns.
5. **Correlate with logs:** Combine firewall logs, PCAPs, DNS logs, and authentication logs to form a complete timeline.

Common Network Attacks Investigated:

- DoS/DDoS
- MITM (Man-in-the-Middle)
- DNS spoofing and cache poisoning
- Port scanning and reconnaissance
- ARP poisoning
- Data exfiltration
- Malware command-and-control communication

Tools for Network Forensics

Here are widely used tools that assist in collecting and analyzing network evidence.

1. **Wireshark:** Graphical tool for live packet capture and deep packet inspection. Shows protocols, payloads, and reconstructs sessions.
2. **tcpdump:** Command-line packet capture tool for Unix/Linux. Useful for quick captures and writing PCAP files.
3. **Tshark:** Command-line version of Wireshark with powerful filtering options.
4. **Network Miner:** A passive network analysis tool that extracts files, cookies, credentials, and metadata from PCAPs.
5. **Nmap:** Used for scanning networks, identifying open ports, detecting OS and services — useful for attack reconstruction.
6. **Snort / Suricata:** IDS/IPS tools that generate logs and alerts for suspicious network activity.
7. **Splunk / ELK:** SIEM tools that centralize and analyze network logs for threat detection.

Writing Computer Forensics Reports

A forensic report is the final and most important deliverable of a digital investigation. It must be clear, accurate, legally valid, and easy for both technical and non-technical audiences to understand. Courts rely heavily on this report, so it must follow strict formatting and documentation standards. **Structure of a Forensics Report**

1. **Title Page:** Case name, case number, investigator name, date, organization.
2. **Introduction & Scope:** Purpose of the investigation, what was requested, and what systems were examined.
3. **Terms & Definitions:** Clarifies technical terminology used in the report.
4. **Case Background:** Description of the incident, complaint, or suspected activity.
5. **Evidence Description:** Details of devices examined, serial numbers, storage sizes, and condition of evidence upon receipt.
6. **Methodology / Procedures:** Step-by-step description of how evidence was acquired, preserved, imaged, and analyzed.
7. **Tools Used:** List of forensic tools with versions and validation details.
8. **Findings / Analysis Results:** Recovered data, logs, timelines, artifacts, deleted files, anomaly detection, etc.
9. **Timeline of Events:** A chronological reconstruction of user actions or attack flow.
10. **Interpretation / Conclusion:** Logical explanation of what the evidence indicates.
11. **Supporting Screenshots, Hash Values & Appendices:** Hashes, logs, screenshots, exported files, and additional details.
12. **Chain of Custody Documentation:** Records of evidence handling.